

PROJECT BOATY

Subsystem Specification Helm

Navigate, enforce the fence and run every native failsafe on an independent RTOS processor, whatever else has failed.

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Parents	SRS Issue D, ADD Issue D, ICD Issue C, FMEA Issue B
Content	37 allocated SRS requirements → 41 subsystem requirements (36 M, 3 S, 2 C)

Document control

Issue	Date	Change	By
A	28 September 2026	First issue, for review.	Claude (drafted)
B	28 September 2026	CR-02: microSD logging (HLM-D02). FMEA actions A-01, A-02, A-06, A-18: HLM-D24 formalised; HLM-D39 to D41 added; parameter baseline extended.	Claude, owner decisions (CR-02, FMEA actions)

Role	Name	Signature / date
Author	Claude, for Project Boaty	
Reviewer / approver	Project owner	

Review guidance: section 5 is what will be built and tested. Check that each requirement is right, sufficient and testable. Section 4 shows every parent requirement is covered (the build script checks this).

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1. Scope

1.1 Purpose

Navigate, enforce the fence and run every native failsafe on an independent RTOS processor, whatever else has failed.

1.2 Boundary

Inside this subsystem	Outside (and who owns it)
• Flight controller hardware (F405-class) • ArduPilot Rover firmware (third party) and the controlled parameter set • GNSS + compass module and mast cabling • Fence and mission storage • Dataflash / SD logging	• Everything that sends commands (MCN via IF-02, MCP via IF-04) • Motor drives (PRP) • Power sources (PWR)

1.3 Breakdown

Item	Description
HLM-1 Flight controller	F405-class, ChibiOS; IMU, baro, UARTs, outputs, ADCs
HLM-2 Firmware	ArduPilot Rover, pinned version
HLM-3 Parameter set	params/boaty-mk1.parm (controlled)
HLM-4 GNSS/compass	u-blox M10 + magnetometer on the mast

2. Applicable documents

Document	Use
BOATY-SRS-001 Issue D	Parent requirements
BOATY-ADD-001 Issue D	Allocation, decisions, budgets, early verification items
BOATY-ICD-001 Issue C	Interface definitions (all listed in section 3 apply in full)
BOATY-FMEA-001 Issue B	Failure modes and actions carried into this issue

3. Interfaces

The subsystem shall meet every interface below as defined in the ICD. 'Owner' means this subsystem maintains the definition.

Interface	Other end	Role	ICD
IF-02 Command & telemetry	Mission Control	Conforms	\$4
IF-04 Companion link	Mission computer	Conforms	\$4
IF-05 Motor commands	Propulsion	Owner	\$6
IF-07 Helm power & sensing	Power	Conforms	\$6
IF-09 Beacon drive	Recovery & signalling	Owner	\$6
IF-19 Mast socket	Hull & structure	Conforms	\$8
IF-22 GNSS & compass	internal	Owner	\$6

4. Parent requirements

Primary (37): this subsystem is responsible for meeting and verifying these SRS requirements.

SRS	Pri	Requirement (abridged)	Met by
MOD-001	M	The system shall implement exactly these boat modes: DISARMED, HOLD, MANUAL, AUTO and RTL, as defined in section 2.3.	HLM-D05
MOD-002	M	On power-up the boat shall enter DISARMED, with both motors inhibited.	HLM-D04
MOD-004	M	The system shall refuse to arm unless every pre-arm check (PRE-001 to PRE-008) passes, and shall state the failing...	HLM-D08
MOD-005	M	When an AUTO mission completes, the boat shall enter RTL. On reaching home (within the waypoint acceptance radius) it...	HLM-D06
MOD-008	C	The boat shall disarm automatically after 60 s in HOLD at home with no operator input, and announce it.	HLM-D07
PRE-001	M	GNSS shall have a 3D fix with at least 8 satellites and HDOP no greater than 1.5.	HLM-D09
PRE-002	M	A geofence shall be loaded and enabled, and home shall lie inside the inclusion fence and outside all exclusion zones.	HLM-D10
PRE-003	M	Battery state of charge shall be at least 80%.	HLM-D11
PRE-004	M	The heading sensor shall report healthy and calibrated.	HLM-D12
PRE-007	M	The independent safety layer (SAF-001) shall report healthy.	HLM-D13
PRE-008	M	Logging shall be active on the helm and Mission Control, each with at least 1 GB free.	HLM-D02, HLM-D14
NAV-002	M	Cruise speed in calm water shall be 1.0 m/s $\pm$ 0.2 m/s.	HLM-D15
NAV-003	M	Speed through the water shall be limited to 1.5 m/s in every mode.	HLM-D16
NAV-004	M	In AUTO the boat shall pass within the acceptance radius of each waypoint. The radius is configurable from 1 to 10 m,...	HLM-D17
NAV-005	S	Cross-track error between waypoints shall be no more than 3 m RMS in wind up to 3 m/s.	HLM-D18
NAV-006	S	In HOLD with station-keeping, the boat shall stay within 5 m of the hold point in wind up to 3 m/s.	HLM-D18
NAV-007	M	In RTL the boat shall navigate to home without entering any exclusion zone or leaving the inclusion fence.	HLM-D19
NAV-008	S	Above 0.3 m/s the heading estimate shall use GNSS course as well as the magnetometer, so a disturbed compass cannot...	HLM-D20
FEN-001	M	The system shall support one inclusion polygon of 3 to 70 vertices and up to 10 exclusion zones, each a polygon or...	HLM-D10, HLM-D21
FEN-004	M	The fence shall be enforced in every armed mode, including MANUAL.	HLM-D22
FEN-005	M	On a fence breach the boat shall enter RTL within 1 s of the breach.	HLM-D23
FEN-006	M	If the boat stays outside the inclusion fence for more than 30 s, or gets more than 10 m outside it, the motors shall...	HLM-D24
FEN-007	M	Fence and exclusion zones shall not be changed while the boat is armed.	HLM-D25
FS-001	M	Low battery: when estimated state of charge falls to 35%, the boat shall enter RTL. At 15% it shall continue RTL at...	HLM-D26, HLM-D40
FS-002	M	Link loss in MANUAL: with no command link for 2 s the boat shall enter HOLD, and after 10 s it shall enter RTL.	HLM-D27
FS-003	M	Link loss in AUTO: the boat shall continue the mission. If the link is still lost after 60 s it shall enter RTL.	HLM-D27
FS-004	M	Position loss: if the GNSS fix is lost, HDOP exceeds 2.5 or the position estimate is unhealthy for 3 s, the motors...	HLM-D28, HLM-D39

SRS	Pri	Requirement (abridged)	Met by
FS-005	M	Stuck detection: if commanded thrust is at least 50% and ground speed stays below 0.1 m/s for 5 s, the boat shall...	HLM-D29
FS-007	M	Mission computer failure: if the helm receives no mission computer heartbeat for 3 s, it shall complete the loaded...	HLM-D30
FS-011	M	When several failsafes are active, the most conservative response shall win: motors stopped over RTL over continuing...	HLM-D31
FS-012	M	Every failsafe event shall be logged, and announced at Mission Control within 2 s if the link is up.	HLM-D32
SAF-001	M	Fence and failsafe functions shall run on a processor separate from the mission computer and Mission Control. That...	HLM-D01, HLM-D02, HLM-D03
SAF-002	M	Those functions shall keep working with the mission computer, Mission Control and the radio link all failed.	HLM-D30, HLM-D33
SAF-003	M	The mission computer and Mission Control shall command the helm only through the helm interface (SWE-002). They shall...	HLM-D34
COM-007	C	An independent manual-override radio link (e.g. ExpressLRS) may be added.	HLM-D37
MEC-013	M	The GNSS and compass shall be mounted at least 150 mm from motor and power wiring.	HLM-D35, HLM-D41
LOG-001	M	The helm shall log position, attitude, mode, commands, battery and failsafe events at 5 Hz or faster.	HLM-D36

Supporting (22): another subsystem owns these, but this subsystem must provide something for them. Where relevant, section 5 includes the contribution.

MOD-003, MOD-006, PRE-005, NAV-001, FS-008, FS-009, FS-010, SAF-007, MIS-001, MIS-002, MIS-003, MIS-004, MIS-005, MIS-006, DET-004, MC-014, REC-004, PWR-005, PWR-006, PWR-010, SWE-008, ENV-002

5. Subsystem requirements

Requirements are numbered HLM-Dnn. Pri, Ver and Stage follow SRS conventions (M/S/C; Test, Demonstration, Inspection, Analysis, Simulation; SIM, BENCH, POOL, LAKE). 'Trace' lists the parent SRS requirements, interfaces (IF), decisions (DD) and early verification items (V) it implements.

5.1 Hardware and firmware

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D01	The helm shall run a pinned ArduPilot Rover stable release (≥ 4.5) built for the chosen board. Version and hash are recorded in the repository.	M	I	BENCH	SAF-001, SWE-006
HLM-D02	The flight controller shall provide: ≥ 2 free UARTs (GNSS, companion), ≥ 3 DShot-capable outputs, ≥ 2 analogue battery inputs, I2C, IMU and baro, and a microSD slot (≥ 8 GB card) for logging (CR-02).	M	I	BENCH	SAF-001, PRE-008, V-01, DD-15
HLM-D03	No project-written code (including Lua) shall run on the flight controller. It runs ChibiOS only.	M	I	BENCH	SAF-001, DD-13
HLM-D04	On power-up the helm shall be disarmed with motor outputs at neutral. Auto-arming shall be disabled.	M	T	BENCH	MOD-002

5.2 Modes

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D05	Only STEERING, HOLD, LOITER, AUTO, RTL and GUIDED (weed-shedding only) shall be used. No RC mode switch is configured.	M	I	SIM	MOD-001, DD-12
HLM-D06	At mission end the helm shall RTL and then hold at home. The mission-done behaviour is set to hold as a backstop.	M	S	SIM	MOD-005, V-04
HLM-D07	Auto-disarm after 60 s at home is performed by Mission Control (MCN-D12). The helm shall accept it.	C	S	SIM	MOD-008

5.3 Pre-arm checks

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D08	All ArduPilot arming checks shall be enabled, and failures reported as STATUSTEXT (MCN translates them).	M	S	SIM	MOD-004
HLM-D09	Arming shall need a 3D fix with HDOP ≤ 1.5 . The ≥ 8 satellite check is done by Mission Control (MCN-D15).	M	T	BENCH	PRE-001
HLM-D10	Arming shall need the fence enabled and loaded, with home inside it.	M	S	SIM	PRE-002, FEN-001
HLM-D11	Arming shall need $\geq 80\%$ battery capacity remaining.	M	S	SIM	PRE-003
HLM-D12	Arming shall need calibrated, consistent compass and EKF.	M	T	BENCH	PRE-004
HLM-D13	Arming shall be refused with motor-rail voltage < 9.0 V (key out).	M	T	BENCH	PRE-007, MOD-003, V-13
HLM-D14	Arming shall need logging available.	M	T	BENCH	PRE-008

5.4 Navigation

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D15	Cruise speed shall be set to 1.0 m/s and tuned to ± 0.2 m/s.	M	T	POOL	NAV-002
HLM-D16	Waypoint and RTL speeds shall be ≤ 1.5 m/s. Maximum throttle shall be limited so full throttle in STEERING gives ≤ 1.5 m/s in calm water.	M	T	POOL	NAV-003
HLM-D17	Waypoint acceptance radius shall default to 3 m (configurable 1-10 m).	M	S	SIM	NAV-004

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D18	Steering and speed controllers shall be tuned to ≤ 3 m RMS cross-track error and ≤ 5 m loiter error in 3 m/s wind.	S	T	LAKE	NAV-005, NAV-006
HLM-D19	RTL shall plan around exclusion zones inside the inclusion fence (fence-aware path planning).	M	S	SIM	NAV-007, V-01
HLM-D20	The heading estimate shall fall back to GNSS-velocity yaw when the compass disagrees above 0.3 m/s.	S	S	SIM	NAV-008, V-16

5.5 Fence

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D21	The helm shall hold one inclusion polygon (≤ 70 vertices) and ≤ 10 exclusion polygons or circles, plus a 100 m circle backstop about home at Milton.	M	S	SIM	FEN-001, OPS-006
HLM-D22	The fence shall be enforced in every armed mode used.	M	S	SIM	FEN-004, V-02
HLM-D23	A breach shall trigger RTL within 1 s.	M	S	SIM	FEN-005
HLM-D24	Persistent breach (> 30 s or > 10 m outside) shall stop the motors. If V-14 finds no native mechanism, Mission Control (MCN-D59) and the MCP (B7) shall each independently command HOLD on that condition, and either suffices. The helm's fence RTL stays the primary response.	M	S	SIM	FEN-006, V-14, A-18, FM-14
HLM-D25	Fence changes while armed shall be refused by every sender (IF-14, IF-04 filter) and by the helm if supported (V-17).	M	S	SIM	FEN-007, V-17

5.6 Failsafes

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D26	Battery failsafe: RTL at 35% remaining; at 15%, continue RTL, alarm and reduce speed (V-15).	M	S	SIM	FS-001, V-15
HLM-D27	GCS failsafe on system-255 heartbeats: 2 s timeout, action HOLD, continue in AUTO. The later RTL steps are done by MCP B4 (MCP-D12).	M	S	SIM	FS-002, FS-003, V-03, V-06
HLM-D28	EKF/position failsafe $\rightarrow$ HOLD (motors stop). Recovery RTL is done by MCP B6 (MCP-D15).	M	S	SIM	FS-004
HLM-D29	Crash/stuck check $\rightarrow$ HOLD, raising an event the MCP can act on.	M	S	POOL	FS-005, V-05
HLM-D30	No companion-computer failsafe shall be configured: the helm completes missions without the MCP.	M	S	SIM	FS-007, SAF-002
HLM-D31	When failsafes combine, motors-stopped outranks RTL, which outranks continue. Demonstrated in SITL combinations.	M	S	SIM	FS-011
HLM-D32	Every failsafe shall be logged and reported by STATUSTEXT.	M	S	SIM	FS-012

5.7 Independence, identity and logging

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D33	All fence and failsafe functions shall work with the MCP, Mission Control and the link all absent.	M	S	SIM	SAF-002
HLM-D34	Only system 255 is the GCS. Parameter writes are accepted only while disarmed through IF-14. Boat services are filtered (IF-04).	M	T	SIM	SAF-003, IF-02, IF-04
HLM-D35	The GNSS/compass shall be mast-mounted ≥ 150 mm from power wiring, with motor interference $\leq 30\%$ after a motor-interference calibration.	M	T	BENCH	MEC-013
HLM-D36	The helm shall log position, attitude, mode, commands, battery and failsafes at ≥ 5 Hz.	M	T	BENCH	LOG-001

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D37	A spare UART shall be kept for an optional ExpressLRS receiver, with RC failsafe = RTL if fitted.	C	I	BENCH	COM-007
HLM-D38	The beacon output shall be configured per IF-09.	S	T	BENCH	IF-09, REC-004

5.8 FMEA-driven (Issue B)

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D39	GNSS glitch rejection (EKF3, glitch radius ≤ 25 m) and GNSS data-timeout detection shall be enabled. A glitch or frozen position shall not produce an uncommanded exit from the fence (SC-20, SC-21, L2-04).	M	S	SIM	FS-004, FS-013, A-01, FM-02, FM-03
HLM-D40	Voltage thresholds shall back up the mAh battery failsafe: low 10.2 V and critical 9.6 V under load for ≥ 10 s, with the same actions. Whichever triggers first applies.	M	T	BENCH	FS-001, A-06, FM-15
HLM-D41	Compass interference shall be measured on rig L2 at full thrust before the hull layout is frozen. Above 30%, wiring or mast height changes before build.	M	T	BENCH	MEC-013, A-02, FM-04

6. Controlled parameter baseline

The baseline configuration that implements the requirements above. **The authoritative list is `params/boaty-mk1.parm` in the repository.** Names and values are confirmed against the flashed firmware version in SITL before the file is frozen (HLM-D01). 'Confirm' shows where each is proven.

Function	Parameter(s)	Baseline	Satisfies	Confirm
Vehicle	FRAME_CLASS	Boat	MOD-001	V-01
Motor outputs	SERVO1_FUNCTION / SERVO2_FUNCTION	ThrottleLeft (73) / ThrottleRight (74)	NAV-001, IF-05	Bench
Output protocol	MOT_PWM_TYPE	DShot300	IF-05	TBC-05
Throttle cap	MOT_THR_MAX	Tuned: full throttle ≤ 1.5 m/s	NAV-003	Pool
Cruise	CRUISE_SPEED / CRUISE_THROTTLE	1.0 m/s / measured	NAV-002	Pool
Waypoints	WP_SPEED / WP_RADIUS	1.0 m/s / 3 m	NAV-004	SITL
RTL	RTL_SPEED	0 (= WP_SPEED)	NAV-007	SITL
Loiter	LOIT_RADIUS	2 m	NAV-006	Lake
Path planning	OA_TYPE	Dijkstra (fence-aware)	NAV-007	V-01
Identity	SYSID_THISMAV / SYSID_MYGCS	1 / 255	IF-02	V-06
Companion port	SERIAL2_PROTOCOL / SERIAL2_BAUD	2 (MAVLink 2) / 115	IF-04	TBC-02
Fence	FENCE_ENABLE / FENCE_TYPE / FENCE_ACTION	1 / polygon + circle / RTL	FEN-001, FEN-005	V-02
Fence backstop	FENCE_RADIUS / FENCE_MARGIN	100 m / 3 m	OPS-006	SITL
Battery	BATT_MONITOR / BATT_CAPACITY	Analogue V+I / measured (≈ 3000 mAh)	PWR-006	Bench
Battery failsafe	BATT_LOW_MAH / BATT_CRT_MAH	35% / 15% of capacity	FS-001	SITL
Battery actions	BATT_FS_LOW_ACT / BATT_FS_CRT_ACT	RTL / RTL	FS-001	V-15
Arming capacity	BATT_ARM_MAH	80% of capacity	PRE-003	SITL
Motor rail	BATT2_MONITOR / BATT2_ARM_VOLT	Analogue V / 9.0 V	MOD-003, PRE-007	V-13
GCS failsafe	FS_GCS_ENABLE / FS_TIMEOUT / FS_ACTION	2 (continue in AUTO) / 2 s / Hold	FS-002, FS-003	V-03
EKF failsafe	FS_EKF_ACTION	Hold	FS-004	SITL
Stuck	FS_CRASH_CHECK	Hold	FS-005	V-05
GNSS	GPS_TYPE / GPS_HDOP_GOOD	Auto / 150	PRE-001	Bench
Arming	ARMING_CHECKS	All	MOD-004	SITL
Heading	COMPASS_ORIENT (+ EKF3 GSF yaw)	Per module / default on	NAV-008	TBC-09, V-16
Beacon	NTF_LED_TYPES / SERVO3_FUNCTION	NeoPixel / NeoPixel1	REC-004, IF-09	Bench
Logging	LOG_BACKEND_TYPE / LOG_BITMASK	File / default + ≥ 5 Hz	LOG-001	Bench
Mission end	MIS_DONE_BEHAVE	Hold (backstop)	MOD-005	V-04

Function	Parameter(s)	Baseline	Satisfies	Confirm
Battery backstop	BATT_LOW_VOLT / BATT_CRT_VOLT	10.2 V / 9.6 V	FS-001 (HLM-D40)	L2-09
GNSS glitch	EK3_GLITCH_RAD	25 m (default; confirm)	FS-004 (HLM-D39)	SC-20

6.1 Where each failsafe step lives

Requirement	Helm (native)	Boat Python (MCP)	Mission Control
FS-001	RTL at 35%; RTL + alarm at 15%	-	Announce; reduce speed if V-15 shows no native way
FS-002	HOLD at 2 s	RTL at 10 s (B4)	Show 'link lost'
FS-003	Continue in AUTO	RTL at 60 s (B4)	-
FS-004	HOLD (motors stop)	RTL after 10 s healthy (B6)	Announce
FS-005/006	HOLD on stuck	≤ 3 astern bursts (B5)	Announce
FEN-005/006	RTL on breach; persistent → stop (V-14)	HOLD if persistent (interim)	HOLD if persistent (interim)

Rule: nothing outside the helm is the last line of defence. Boat Python and Mission Control only add later, safer steps.

7. Design constraints and budgets

- ADD DD-02: F405-class, subject to V-01
- ADD DD-13: no Lua; Python on the MCP covers Lua-type behaviours
- No project-written code on the FC (SAF-001)

Budget	Allocation
Mass allocation	≤ 40 g (estimate 32 g)
Cost allocation	£49 (FC with microSD £35 est., GNSS £14)
Power	≈ 0.8 W including GNSS

8. Verification and open items

Each requirement is verified at its stage using the method shown. Test procedures are written per stage and reference these IDs.

Stage	Test	Demo	Inspect	Analysis	Sim	Reqs
SIM	1	-	1	-	21	23
BENCH	10	-	4	-	-	14
POOL	2	-	-	-	1	3
LAKE	1	-	-	-	-	1

Ref	Open item
V-01	Firmware feature check for the chosen board, incl. path planning around fences
V-02 ... V-06	Mode, failsafe and identity behaviours in SITL
V-14	Native mechanism for FEN-006 (persistent breach → motors stop)
V-15	Reduced RTL speed on critical battery (FS-001)
V-16	GNSS-velocity yaw fallback behaviour (NAV-008)
V-17	Whether the helm itself refuses fence changes while armed
HLM-OI-1 / CR-02	Closed: owner chose a flight controller with microSD (ADD DD-15)
A-18	Interim persistent-breach path formalised in HLM-D24 until V-14 is resolved